EEL1010
Introduction to Electrical Engineering

Diodes

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Diodes

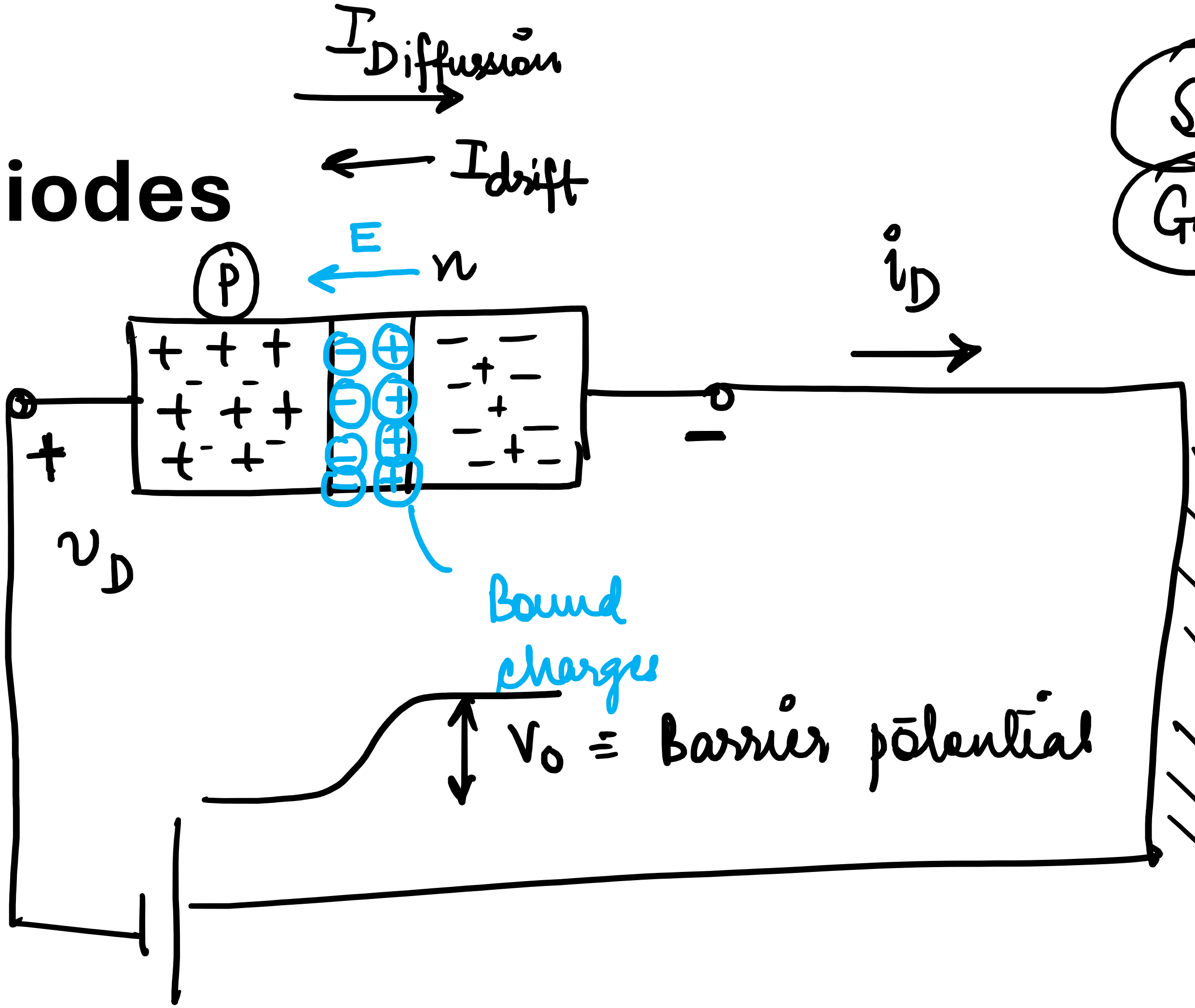
Compound Semiconductors : $\boxed{\text{Light-based Application}}$

Group 3	Group 5
Al	P
Ga	As
In	

GaAs

GaAsP

Diodes



$\boxed{E_g}$

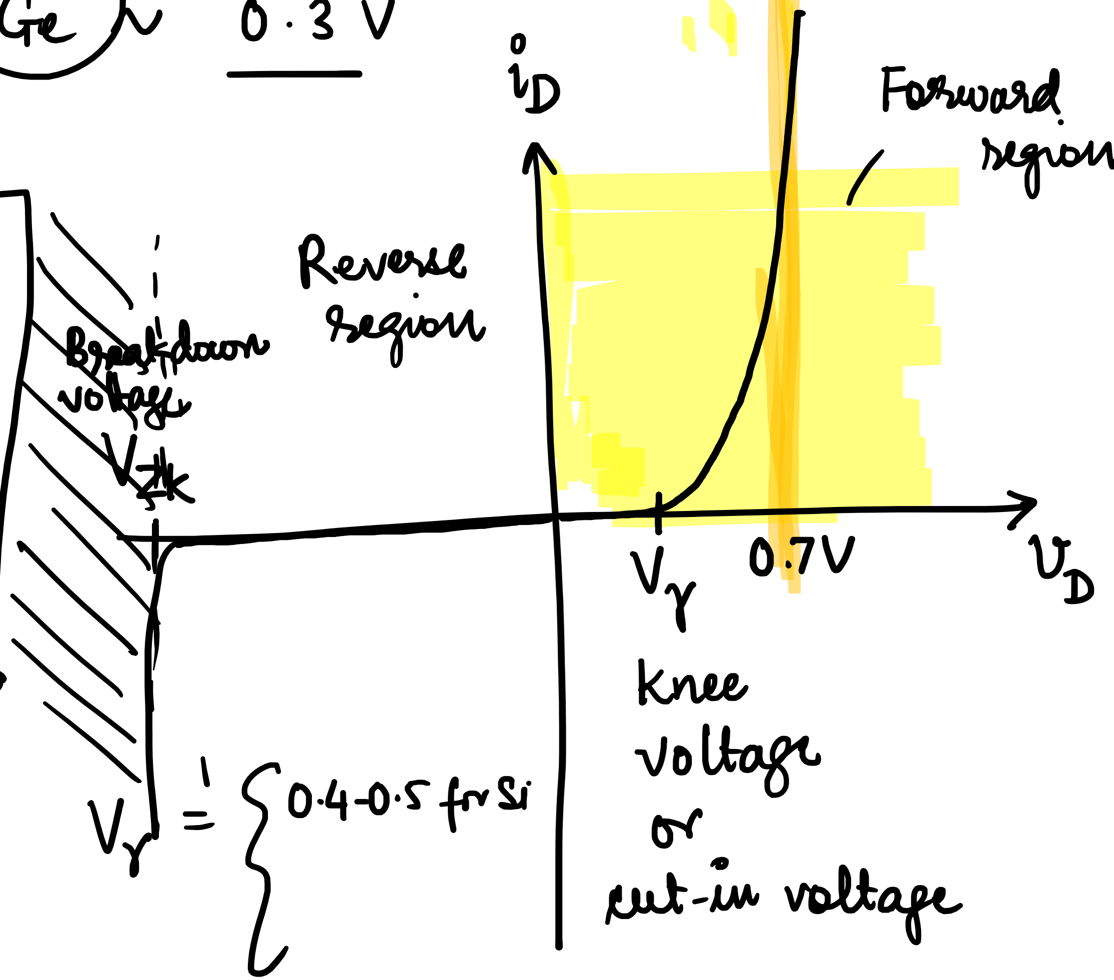
v_D : pot. of p wrt n

i_D : current from p to n

Si ~ $\underline{0.7\,V}$

Ge ~ $\underline{0.3\,V}$

$\boxed{\text{I-V JD}}$



$V_\gamma \doteq \{ 0.4\text{-}0.5 \text{ for Si}$

LEDs

An LED produces light through **electroluminescence**, where electrons recombine with holes in a semiconductor material to emit photons.

The color of the light is determined by the **bandgap energy** of the semiconductor material used.

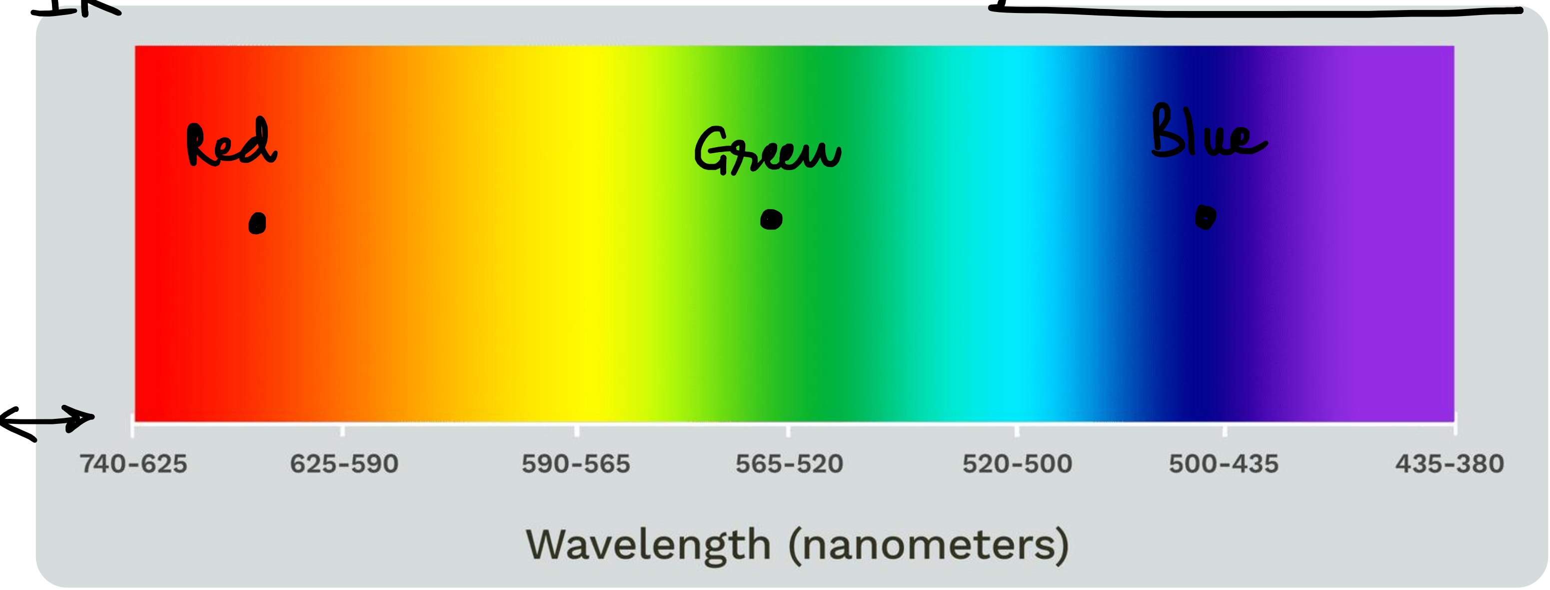
LEDs

(eV)

$$E_g = \frac{hc}{\lambda}$$

$$\boxed{E\ (\text{eV}) = \frac{1240}{\lambda\ (\text{nm})}}$$

IR



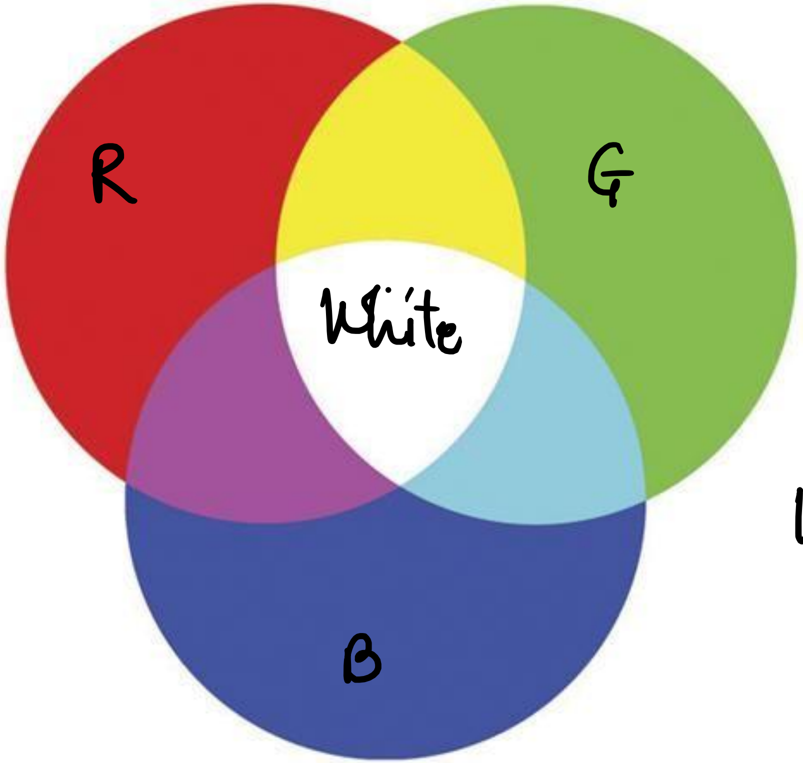
Element	E_g (eV)	λ (nm)	Color
Ge	0.7		X
Si	1.1		X
GaAs	1.4	900	IR
GaAsP	1.9-2.0	~620	Red
⋮			
GaP	2.3	530	Green
GaN + InGaN + AlGaN	2.75	450	Blue
GaN	3.4	360	UV

LEDs

Light

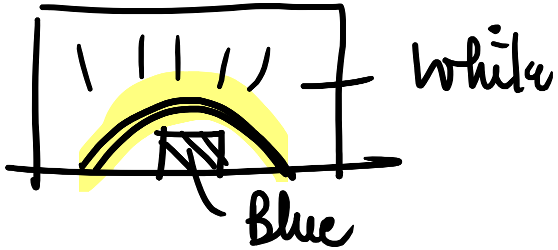
Yellow Phosphor + Blue LED

= white light



Additive color mixing

White



Nobel Prize in Physics 2014

Isamu Akasaki

Hiroshi Amano

Shuji Nakamura

The Nobel Prize in Physics 2014 was awarded jointly to Isamu Akasaki, Hiroshi Amano and Shuji Nakamura "for the invention of efficient blue light-emitting diodes which has enabled bright and energy-saving white light sources"

LED	CFL	Incandescent
Avg Life: 25,000 Hrs	Avg Life: 8,000 Hrs	Avg Life: 1,200 Hrs
No Mercury	Mercury	No Mercury
6-8 Watts	13-15 Watts	60 Watts
Uses 84% less energy	Uses 75% less energy	90% energy lost to heat